

Quantum-Inspired MRFO and DMO — Results as Exact Values

Companion to results_summary.pdf · makespan in seconds instead of percentage gaps

What the numbers are. Makespan is the time, in seconds, at which the last task finishes; lower is better. Task lengths are in MI (million instructions) and VM speeds in MIPS. The lower bound is the theoretical minimum makespan of a problem: no schedule can finish earlier. The percentage gaps in the summary are (makespan – lower bound) / lower bound.

Units. Makespan, lower bound and standard deviation: **seconds (s)**. Task length: **MI** (million instructions). VM speed: **MIPS** (million instructions per second), so a task's execution time = length ÷ speed is in seconds. Migrations: **number of running tasks** moved to another VM per change. λ (migration price) has no unit.

Table 1 · DMO and MRFO, original vs quantum-inspired: makespan in seconds (s) (V4 study)

Algorithm (makespan in s)	30 tasks 5 VMs uniform · high	50 tasks 10 VMs uniform · none	50 tasks 10 VMs bimodal · high	60 tasks 8 VMs bimodal · none	100 tasks 10 VMs uniform · high	100 tasks 10 VMs lognormal · low	200 tasks 20 VMs uniform · high
DMO (original)	45.82 ± 0.38 best 44.94	32.94 ± 0.58 best 31.94	35.36 ± 2.62 best 31.24	62.71 ± 1.23 best 59.86	82.54 ± 5.44 best 71.94	44.79 ± 0.89 best 42.23	210.58 ± 20.94 best 169.27
QI-DMO (quantum-inspired)	45.26 ± 0.22 best 44.81	30.42 ± 0.39 best 29.67	29.64 ± 1.61 best 26.68	57.82 ± 1.59 best 55.49	48.93 ± 0.24 best 48.44	39.90 ± 0.30 best 39.46	66.32 ± 1.23 best 64.25
MRFO (original)	47.43 ± 1.18 best 45.02	33.41 ± 1.19 best 31.47	35.69 ± 3.01 best 30.65	64.39 ± 2.40 best 60.72	86.57 ± 12.05 best 58.82	43.96 ± 1.51 best 39.77	152.08 ± 11.83 best 131.87
QI-MRFO (quantum-inspired)	45.00 ± 0.21 best 44.71	29.28 ± 0.23 best 28.66	28.12 ± 1.91 best 25.60	57.53 ± 1.55 best 55.03	47.59 ± 0.24 best 47.31	39.09 ± 0.19 best 38.91	62.53 ± 0.56 best 61.90
Genetic Algorithm	44.91 ± 0.16 best 44.64	29.39 ± 0.22 best 29.04	28.51 ± 1.53 best 26.38	57.19 ± 1.52 best 55.03	47.74 ± 0.18 best 47.49	39.09 ± 0.17 best 38.85	62.81 ± 0.38 best 62.18
PSO	45.87 ± 0.82 best 44.98	30.22 ± 0.50 best 29.37	28.64 ± 1.61 best 26.22	59.10 ± 2.41 best 55.03	62.99 ± 6.95 best 50.25	40.38 ± 0.83 best 39.48	108.38 ± 16.11 best 84.27
Max-Min heuristic	45.46 ± 0.00 best 45.46	28.98 ± 0.00 best 28.98	27.06 ± 0.00 best 27.06	55.03 ± 0.00 best 55.03	47.45 ± 0.00 best 47.45	38.98 ± 0.00 best 38.98	61.33 ± 0.00 best 61.33
Min-Min heuristic	48.18 ± 0.00 best 48.18	32.29 ± 0.00 best 32.29	37.35 ± 0.00 best 37.35	74.07 ± 0.00 best 74.07	50.87 ± 0.00 best 50.87	54.60 ± 0.00 best 54.60	66.63 ± 0.00 best 66.63
Random search	47.54 ± 0.84 best 45.65	32.71 ± 0.67 best 31.08	33.77 ± 0.88 best 31.51	63.44 ± 1.47 best 60.46	91.93 ± 6.33 best 74.52	44.41 ± 0.73 best 42.78	158.41 ± 6.23 best 144.72
Lower bound (s), theoretical minimum	44.53	28.39	25.11	52.94	47.06	38.68	60.83

Each cell: mean ± standard deviation over 30 independent runs of 20 000 evaluations, in seconds; below it, the best of the 30 runs, in seconds. Column headings: tasks, VMs, task-length distribution · VM speed heterogeneity (high = 250–2000 MIPS, low = 900–1100 MIPS, none = all 1000 MIPS). Max-Min and Min-Min are deterministic, so their standard deviation is 0. Source: results/baseline_full.csv.

Table 2 · Adding the swap move: makespan in seconds (s) on 80 unseen problems (V5 held-out test)

Problem type (10 problems each)	Lower bound (s)	MRFO original (s)	QI-MRFO no swap (s)	QI-MRFO + swap ours (s)	Genetic Algorithm (s)	GA + swap (s)	Max-Min heuristic (s)
60 tasks, 12 VMs, uniform · low	27.78	34.03	28.94	27.85	28.90	28.68	28.23
80 tasks, 8 VMs, uniform · high	49.26	71.65	49.77	49.29	49.86	49.71	49.75
100 tasks, 10 VMs, bimodal · high	72.91	108.03	75.50	73.01	75.21	73.51	73.10
100 tasks, 20 VMs, lognormal · low	39.12	43.43	41.05	40.19	40.51	39.86	39.12
120 tasks, 12 VMs, lognormal · high	49.11	73.66	49.75	49.15	49.97	49.86	49.25
150 tasks, 15 VMs, uniform · high	54.21	105.07	55.18	54.29	55.38	55.38	54.72
200 tasks, 10 VMs, bimodal · none	146.49	165.89	147.24	146.49	147.03	146.88	146.68
300 tasks, 30 VMs, uniform · high	53.31	142.26	58.12	53.48	56.41	56.07	53.74

Each cell: mean makespan over the 10 problems of that type (each problem averaged over its 2 runs; the heuristic is deterministic). The lower bound here is the tighter preemptive bound.
Source: results/h5_test/records.csv.

Table 3 · Changing cloud, against the GA: makespan in seconds (s) after each change (V5)

Change type (100 tasks, 10 VMs unless stated)	Lower bound (s)	QI-MRFO + swap, ours (s)	GA + swap (s)	QI-MRFO better in (runs)
Task churn (20 % replaced)	54.75	54.88	56.57	10 of 10
VM speed drift	46.83	46.96	48.34	10 of 10
VM failure	127.05	127.09	127.40	10 of 10
VM added	38.42	38.83	42.16	10 of 10
Mixed events	53.05	53.19	54.90	10 of 10
Mixed, 200 tasks, 20 VMs, heavy-tailed	47.35	47.74	50.05	10 of 10

Each cell: mean over the 8 changes of a scenario and 10 independent runs. Both methods use the swap move and keep their state between changes. Source: results/h6_dynamic (makespans recomputed from the stored gaps and the regenerated lower bounds; exact to rounding).

Table 4 · Changing cloud with a cost for moving running tasks: makespan (s) and migrations (tasks per change) (V5)

Best heuristic = the cheaper of full Max-Min recompute and zero-migration repair. Each cell: mean over the 8 changes and 10 runs. Tasks moved = number of running tasks migrated to another VM at each change (a count, unit: tasks per change). The price λ says how much one migration costs relative to makespan: the methods optimise makespan $\times (1 + \lambda \times \text{migrations} / \text{movable tasks})$, so at a low price extra migrations are worth a shorter makespan and at a high price they are not. Source: results/h13_event_gamma.

Low migration price ($\lambda = 0.05$)

Change type	Lower bound (s)	Makespan (s): best heuristic	Makespan (s): QI-MRFO (ours)	Tasks moved: best heuristic	Tasks moved: QI-MRFO (ours)
Task churn (20 % replaced)	53.28	54.23	53.47	2.5	7.5
VM speed drift	52.01	52.42	52.21	83.5	47.7
VM failure	122.35	123.11	122.49	3.0	4.7
VM added	37.72	38.37	38.36	62.5	41.2
Mixed events	56.32	57.12	56.56	49.1	31.1
Mixed, 200 tasks, 20 VMs, heavy-tailed	43.06	43.35	43.98	134.5	52.9

Medium migration price ($\lambda = 0.2$)

Change type	Lower bound (s)	Makespan (s): best heuristic	Makespan (s): QI-MRFO (ours)	Tasks moved: best heuristic	Tasks moved: QI-MRFO (ours)
Task churn (20 % replaced)	53.28	54.40	53.68	0.0	2.3
VM speed drift	52.01	52.50	52.56	82.5	32.4
VM failure	122.35	123.19	122.61	0.0	1.4
VM added	37.72	40.89	40.06	25.1	22.2
Mixed events	56.32	57.98	57.07	31.5	18.3
Mixed, 200 tasks, 20 VMs, heavy-tailed	43.06	44.78	44.88	72.4	31.8

High migration price ($\lambda = 1.0$)

Change type	Lower bound (s)	Makespan (s): best heuristic	Makespan (s): QI-MRFO (ours)	Tasks moved: best heuristic	Tasks moved: QI-MRFO (ours)
Task churn (20 % replaced)	53.28	54.40	53.92	0.0	0.3
VM speed drift	52.01	68.26	55.39	41.5	14.3
VM failure	122.35	123.19	122.80	0.0	0.1
VM added	37.72	51.86	47.24	3.5	7.9
Mixed events	56.32	72.84	59.43	8.4	7.4
Mixed, 200 tasks, 20 VMs, heavy-tailed	43.06	57.25	47.28	30.2	13.2